AIORI HACKATHON 3 | A3-PS004-TC456

Authenticated Gateway-Assisted DNSSEC for IoT Clients

Lightweight security offloading for resource-constrained IoT devices

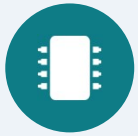
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ABSTRACT

The Problem: IoT Devices Skip DNSSEC



Extreme Hardware Constraints

CCTV cameras, smart meters and sensors have weak CPUs, minimal RAM and strict power budgets.



Heavy Cryptographic Overhead

Standard DNSSEC needs CPU-intensive math to verify DNSKEY public keys and RRSIG signature chains.



Critical Security Risk

Devices bypass DNSSEC to save resources, exposing them to DNS spoofing, cache poisoning and fake update-server redirection.



No Client Authentication

Traditional gateways accept DNS queries blindly, never checking whether the requesting device is authorized.

BACKGROUND

Context and Technical Necessity



DNS vs. DNSSEC

DNS maps names such as update.cctv-vendor.com to IP addresses but has no built-in identity verification. DNSSEC adds a "digital wax seal" that guarantees response authenticity.



The IoT Vulnerability Gap

Smart devices constantly contact cloud servers for firmware updates. An intercepted DNS query lets attackers push malicious payloads straight to the device.



Proposed Optimization

Shift cryptographic validation off the weak device and onto a dedicated, authenticated local Edge Gateway.

LITERATURE & RFC REVIEW

Standards and Existing Studies

RFC 4033 / 4034 / 4035

DNSSEC Specification

Core DNSSEC framework, signature records (RRSIG) and validation chains.

RFC 8499

DNS Terminology

Details stub-resolver limitations on embedded systems.

RFC 9102

Authentication Chain Extensions

Explores offloading validation to network middleboxes.



Existing Solution Gap

Local resolvers such as Pi-hole and Unbound do not perform client-level token authentication, so spoofed internal nodes can misuse the gateway's lookup service.

PROPOSED SOLUTION

Detailed Solution Workflow



STEP 1

Token-Based Authentication

- Device sends request with an X-Device-Token header
- Gateway checks identity against devices.json
- Unauthenticated queries are dropped instantly



STEP 2

Gateway-Assisted DNSSEC

- Requests DNSSEC records (do=1) from upstream resolvers such as 1.1.1.1 / 8.8.8.8
- Verifies signatures via AD flag check and RRSIG verification



STEP 3

Verdict & Admin Alert

- Valid: gateway forwards the verified IP to the device
- Tampered / BOGUS: resolution blocked and a real-time alert sent to the Admin Dashboard

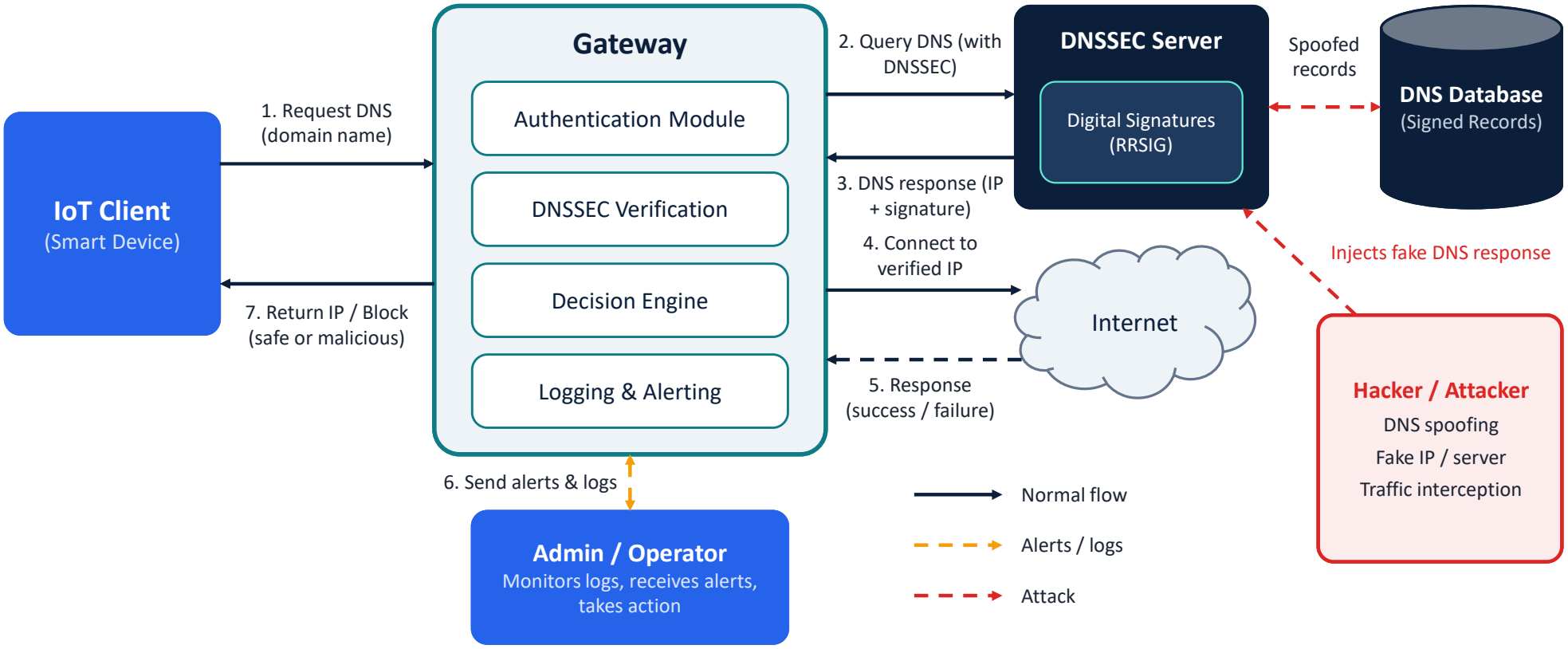
OPTIMIZATION PROPOSED

Traditional vs. Authenticated Gateway

Metric	Traditional IoT DNS	Proposed Gateway Approach
IoT CPU & Memory Usage	✗ High (or DNSSEC disabled)	✓ **Minimal (standard HTTP/REST query)**
DNSSEC Protection	✗ None (disabled to save battery/RAM)	✓ **100% cryptographically verified**
Client Access Control	✗ None (accepts any local query)	✓ **Enforced (X-Device-Token auth)**
Attack Detection	✗ Silent device compromise	✓ **Instant Admin Dashboard alerting**
Battery Life Impact	✗ Heavy battery drain	✓ **Optimized for long device lifespan**

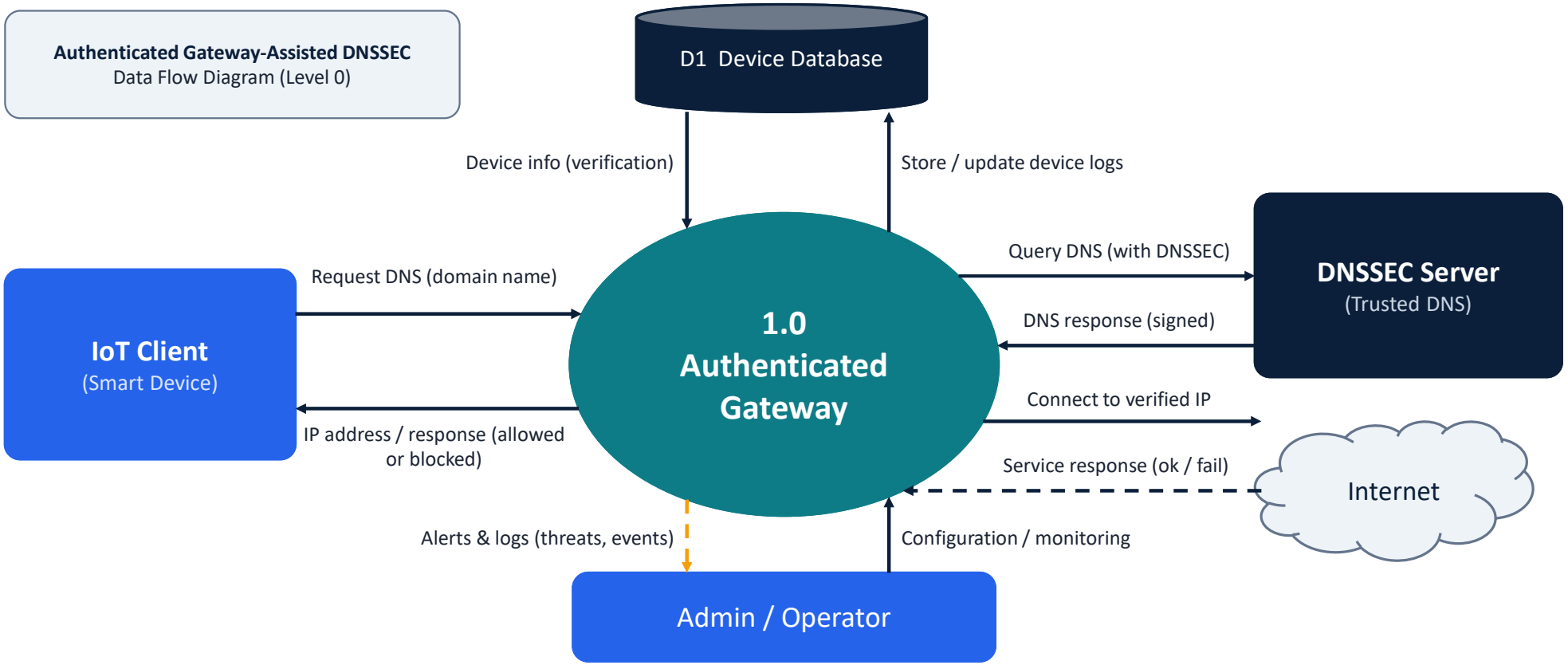
SOLUTION ARCHITECTURE & DESIGN

Architecture Design (UML)



SOLUTION ARCHITECTURE & DESIGN

Data Flow Diagram (Level 0)



DEPLOYMENT TIMELINE

Project Implementation Milestones

1 — 2 — 3 — 4

WEEK 1

Phase 1: Requirements & Protocol Definition

API specs, token schema definition and RFC review.

WEEK 2

Phase 2: Gateway Core Development

FastAPI setup, auth.py implementation and dnspython integration.

WEEK 3

Phase 3: IoT & Attack Simulators

Build iot_device.py client and attack_simulator.py (spoofing tests).

WEEK 4

Phase 4: Dashboard & Integration Testing

Web UI, end-to-end integration and presentation polish.

BIBLIOGRAPHY

References

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Thank You for Your Attention

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